Electrical & Electronic Properties – Part 2

MO-201 Materials Science (Odd Sem 2026-27)



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Reference Textbooks
Materials Science & Engineering by V. Raghavan
Fundamentals of Materials Science & Engineering by W.D. Callister

Electrical & Electronic Properties

- **The Energy Gap in Solids**

- The forbidden gap disappears, if the bottom line of the gap for [110] direction lies above the top line of the gap in the [100] direction

- Such a situation will arise, when the potential energy of an electron is not a strong function of location in a crystal

- The energy bands on either side of a critical value of k are said to overlap and there is no forbidden energy

- Figure 11 is the E–k plot for unidirectional motion of electrons; when all directions of electron motion and all possible reflections are considered together, the ranges of k that lie between the forbidden gaps are called **Brillouin zones**

- The first Brillouin zone contains all the energy levels up to the first E–k discontinuity, the second zone contains all levels between the first and the second discontinuities and so on

- For example, the first zone for an FCC crystal is a polyhedra bounded by planar surfaces of the {111} and {200} types, these two corresponding to the first and the second allowed reflections for the FCC lattice

Electrical & Electronic Properties

- **The Energy Gap in Solids**
- The outermost energy band that is full or partially filled is called the **valence band**
- The band that is above the valence band and that is empty at 0 K is called the **conduction band**
- Solids are classified on the basis of their band structure as metals, semiconductors and insulators
- Metals are those solids which have vacant electron energy states immediately above the highest filled level of the valence band; this can happen in two ways

Either the valence band is only half-filled or a full valence band overlaps the conduction band so that there is no forbidden gap

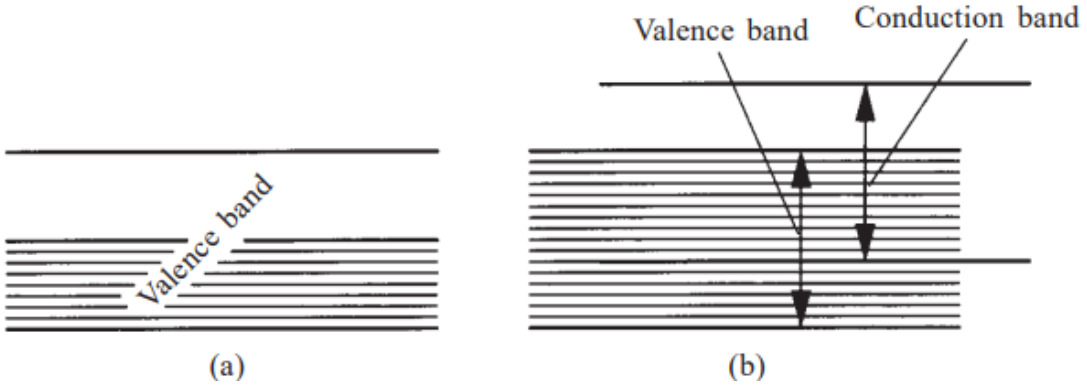


Fig. 13 Metals have partially filled or overlapping bands

Electrical & Electronic Properties

- **The Energy Gap in Solids**

- Monovalent metals such as Cu, Ag and Au have one electron in the outermost orbital; correspondingly, the outermost energy band is only half full in these metals

- Divalent metals such as Mg and Be have overlapping conduction and valence bands; therefore, they also conduct even if the valence band is full

- The band structure of trivalent metals such as Al is similar to that of monovalent metals, in that the outermost band is half full

- In elements of the fourth column, the electrons in the outermost orbital are even in number, as in the divalent metals; the valence band is full, but there is no overlap of the valence band with the conduction band here

Electrical & Electronic Properties

- **The Energy Gap in Solids**

- Taking the case of covalent diamond (an insulator), the potential energy of electrons is a strong function of their location in the crystal; correspondingly, a forbidden gap exists and it is relatively large in magnitude (5.4 eV)

- As we move down the fourth column to the elements below diamond, the electrons of the outermost orbital are farther away from the nucleus; the effect of this increasing separation is dominant over the effect of the increasing charge on the nucleus, so that the electrons are less tightly bound to the nucleus, as we go down the column

- The potential energy of the electron is no longer a strong function of its location; this reduces the energy gap from 5.4 eV in diamond to 1.1 eV in Si, 0.7 eV in Ge, and a mere 0.08 eV in gray Sn; in Pb, the forbidden gap is zero

Electrical & Electronic Properties

- **The Energy Gap in Solids**

- Those materials which have an energy gap of about 2–3 eV or less are called **semiconductors** and those with a gap of more than 3 eV are known as **insulators**

- In contrast to metals, both insulators and semiconductors have a finite forbidden energy gap, the only difference between them arising from the magnitude of the energy gap

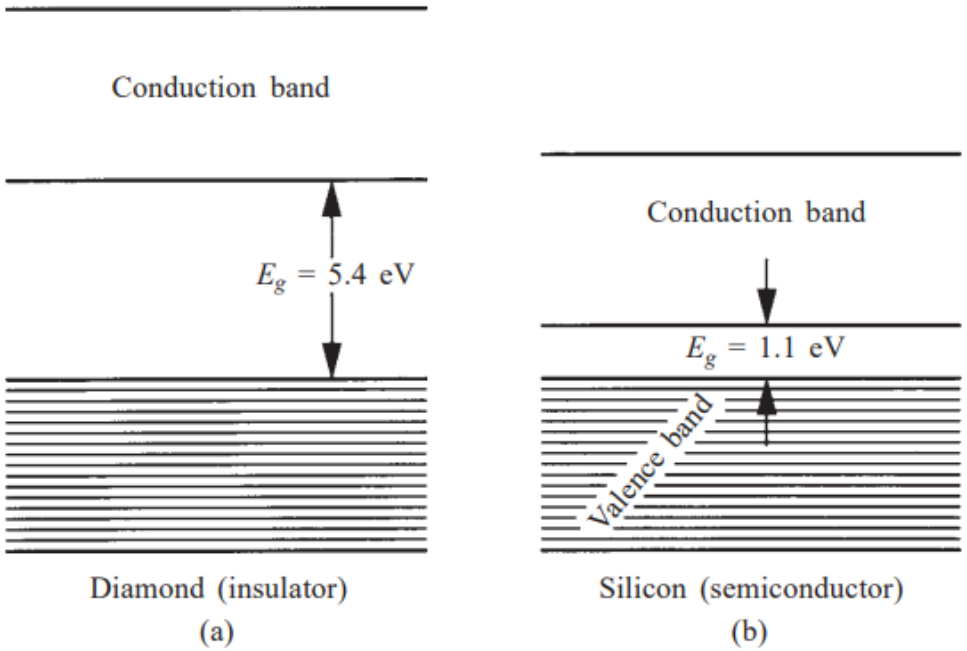


Fig. 14 The difference between (a) an insulator and (b) a semiconductor is in the magnitude of the energy gap

Electrical & Electronic Properties

➢ Intrinsic Semiconductors

- In intrinsic semiconductors, the conduction is due to the intrinsic processes characteristic of the crystal in question, without the influence of impurities
- A pure Si/Ge crystal is an intrinsic semiconductor; the electrons that are excited from the top of the valence band to the bottom of the conduction band by thermal energy are responsible for conduction
- The number of electrons excited across the gap can be calculated from the Fermi–Dirac probability distribution

$$P(E) = \frac{1}{1 + \exp\left[(E - E_F)/kT\right]}$$

- $E - E_F = E_g/2 = 0.55$ eV for Si, which is more than twenty times larger than the thermal energy kT at room temperature (= 0.026 eV)

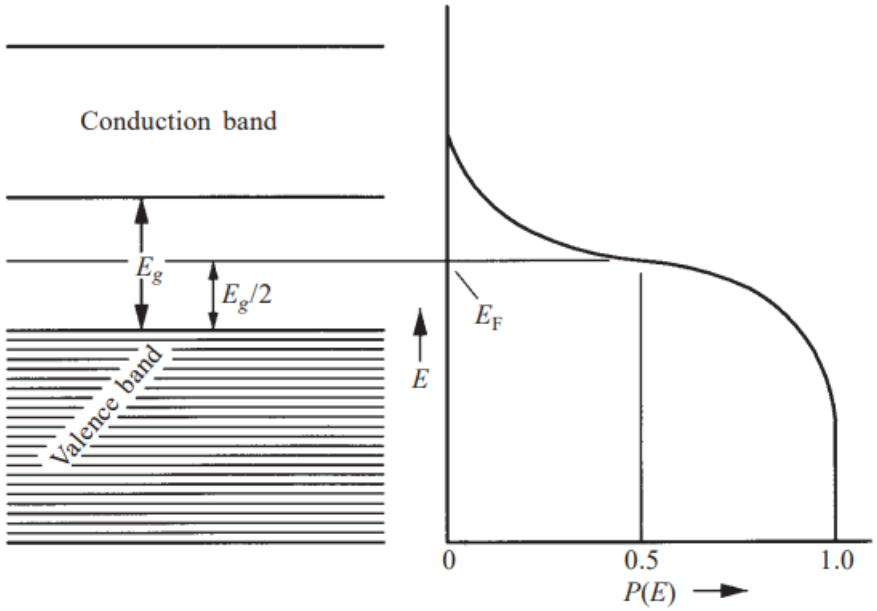


Fig. 15 The Fermi level in an intrinsic semiconductor lies in the middle of the energy gap

Electrical & Electronic Properties

- **Intrinsic Semiconductors**

- Therefore, the factor unity in the denominator can be ignored, in comparison to the exponential term and the probability P(E) of an electron occupying energy level E is

$$P(E) = \exp\left[\frac{-E_g}{2kT}\right]$$

- The fraction of electrons at energy E is equal to the probability P(E); we can then write for the number n of electrons promoted across the gap

$$n = N \exp\left[\frac{-E_g}{2kT}\right]$$

where N is the number of electrons available for excitation from the top of the valence band

- The promotion of some of the electrons across the gap leaves some vacant electron sites in the valence band – these are called **holes**

- As each excited electron leaves back one hole, an intrinsic semiconductor contains an equal number of holes in the valence band and electrons in the conduction band, that is, $n_e = n_h$ (both given by above equation)

Electrical & Electronic Properties

- **Intrinsic Semiconductors**

- Under an externally applied field, the electrons, which are excited into the conduction band by thermal means, can accelerate using the vacant states available in the conduction band

- At the same time, the holes in the valence band also move, but in a direction opposite to that of the electrons

- The conductivity of the intrinsic semiconductor depends on the concentration of these charge carriers, n_e and n_h

- In the case of metals, we referred to the drift velocity acquired by the free electrons in an applied field; similarly, we can define the mobility of conduction electrons and holes μ_e and μ_h as the drift velocity acquired by them under unit field gradient

- Then, conductivity of a semiconductor can be written as

$$\sigma = n_e e \mu_e + n_h e \mu_h$$

where e is the electronic charge and n_e and n_h are concentrations per unit vol. (μ_e and μ_h for Si are 0.14 & 0.05 $m^2V^{-1}s^{-1}$, and 0.39 & 0.19 for Ge respectively)

Electrical & Electronic Properties

- **Intrinsic Semiconductors**

- For an intrinsic semiconductor, the number of charge carriers (electrons and holes) is dependent on temperature in an exponential way and, therefore, increases very rapidly with increasing temperature

- So, the conductivity also increases in a similar fashion with temperature

- The small temperature dependence of mobilities can be neglected, as compared to the effect of temperature on the number of charge carriers

- So, a plot of the log σ vs 1/T (in kelvin) yields a straight line and the energy gap can be computed from the slope of this straight line

Electrical & Electronic Properties

- **Extrinsic Semiconductors**

- In extrinsic semiconductors, the conduction is due to the presence of extraneous impurities

- The process of deliberate addition of controlled quantities of impurities to a pure semiconductor is called **doping;** doping markedly ↑ conductivity of a semiconductor

- Consider a Si crystal which is doped with a fifth column element such as P, As or Sb

- The fifth column element (P in this case) substitutes for a Si atom in the diamond cubic structure

- 4 of the 5 electrons in the outermost orbital of the phosphorus atom take part in the tetrahedral bonding with the four Si neighbours

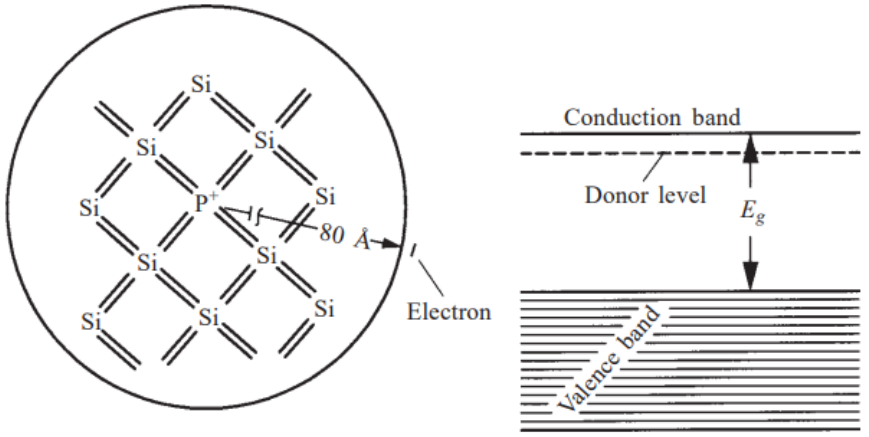


Fig. 16 Silicon doped with phosphorus from the fifth column of the periodic table becomes an n-type semiconductor.

Electrical & Electronic Properties

- **Extrinsic Semiconductors**

- The fifth electron cannot take part in the discrete covalent bonding – it is loosely bound to the parent atom

- It is possible to calculate an orbit for the fifth electron assuming that it revolves around the positively charged phosphorus ion, in the same way as for the 1s electron around the hydrogen nucleus

- There is, however, one important difference; the electron of the phosphorus atom is moving in the electric field of the silicon crystal and not in free space, as is the case in the hydrogen atom

- This brings in the dielectric constant of the crystal into the orbital calculations, and the radius of the electron orbit here turns out to be very large, about 80 Å as against 0.5 Å for the hydrogen orbit

- Such a large orbit evidently means that the fifth electron is almost free and is at an energy level close to the conduction band

Electrical & Electronic Properties

- **Extrinsic Semiconductors**

- Excitation of the fifth electron into the conduction band takes place much more readily than excitation from the valence band of the silicon crystal

- The phosphorus atom is said to donate its electron to the semiconductor

- The energy level of the fifth electron is called the **donor level** (Fig. 16)

- As the elements to the right of the fourth column donate negative charges (electrons), the semiconductors doped with them are called **n-type semiconductors**

- The energy required to excite the fifth electron into the conduction band is known as the **ionization energy**

Ionization Energies for Some Elements in Silicon and Germanium (eV)

Impurity		Silicon	Germanium
n-type	Phosphorus	0.044	0.012
	Arsenic	0.049	0.013
	Antimony	0.039	0.010
p-type	Boron	0.045	0.010
	Aluminium	0.057	0.010
	Gallium	0.065	0.011
	Indium	0.16	0.011

Electrical & Electronic Properties

- **Extrinsic Semiconductors**

- As compared to the energy gap, the ionization energy of an impurity atom is very small; so, at room temperature, a large fraction of the donor level electrons are excited into the conduction band

- This fraction is much larger than the fraction of electrons excited due to the intrinsic process, that is, from the valence band

- According to the law of mass action, the product of the number of electrons in the conduction band and the number of holes in the valence band must be constant

- This condition drastically reduces the number of holes in the n-type semiconductor

- The electrons in the conduction band become the majority charge carriers

- Consider the alternative process of doping a Si crystal with a third column element such as Al, Ga or In

- Al has 3 electrons in the outer orbital; while substituting for Si in the crystal, it needs an extra electron to complete the tetrahedral arrangement of bonds around it

Electrical & Electronic Properties

➢ Extrinsic Semiconductors

- The extra electron can come only from one of the neighbouring Si atoms, thereby creating a vacant electron site (hole) on the Si

- The Al atom with the extra electron becomes a -ve charge and the hole with a +ve charge can be considered to revolve around the Al atom, leading to the same orbital calculations as before

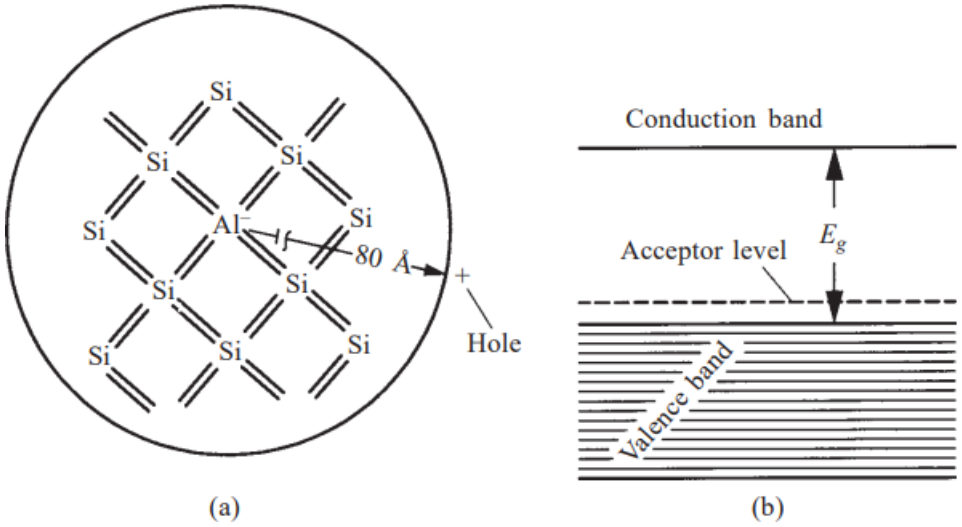


Fig. 17 Silicon doped with the third column element aluminium becomes a p-type semiconductor

Electrical & Electronic Properties

- **Extrinsic Semiconductors**

- The extra electron can come only from one of the neighbouring Si atoms, thereby creating a vacant electron site (hole) on the Si

- The Al atom with the extra electron becomes a -ve charge and the hole with a +ve charge can be considered to revolve around the Al atom, leading to the same orbital calculations as before

- At 0 K, the holes remain bound to the impurity atoms; as the temperature is raised, the holes break away from the impurity atoms due to thermal excitation and become available for conduction

- The ionization energy to free a hole bound to the impurity is approximately the same as the ionization energy of donor electrons in the same crystal

- The bound-hole levels are called **acceptor levels** (Al accepts an electron) and are just above the valence band

Electrical & Electronic Properties

- **Extrinsic Semiconductors**

- The fraction of holes created by thermal excitation in doped Si is much larger than those created by excitation of electrons into the conduction band

- The law of mass action holds good here also and the positive holes become the majority charge carriers

- Such an extrinsic semiconductor is called a **p-type semiconductor**

$$\sigma = n_e e \mu_e + n_h e \mu_h$$

- The above equation describes the conductivity of extrinsic semiconductors as well

- The only difference is that the number of electrons in the conduction band and the number of holes in the valence band are not equal in the case of an extrinsic semiconductor

- One of the two dominates, depending on the type of the extrinsic process

Electrical & Electronic Properties

➢ Extrinsic Semiconductors

- A typical plot of the logarithm of conductivity against temperature is shown for an extrinsic semiconductor in Fig. 18
- The plot can be divided into three distinct regions
- In the low temperature range (corresponding to large values of 1/T), the conductivity is an exponential function of temperature
- The electrons excited from the donor level to the conduction band (or the holes created by excitation to the acceptor level) become available in increasing numbers with increasing temperature
- So, this region is called the extrinsic region

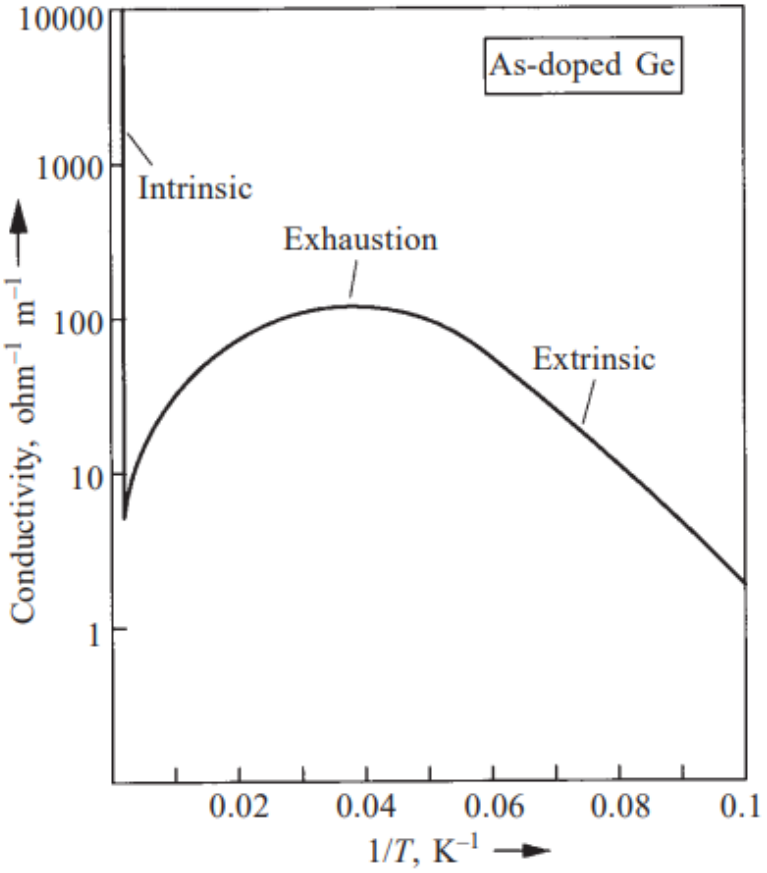


Fig. 18 The variation of log of conductivity with the reciprocal of temperature for an extrinsic semiconductor

Electrical & Electronic Properties

- **Extrinsic Semiconductors**

- From the slope of the straight line in this region, the ionization energy of the impurity can be calculated

- At higher temperatures, there is the second region called the exhaustion region

- Here, the excitation of charge carriers due to impurities is nearing completion, due to the exhaustion of unexcited impurity e^-/h^+

- This is indicated by the flat region of almost zero slope; in this region, the slope can even become positive due to the dominance of the T dependent mobility term in the conductivity equation

- ↑ T results in ↓ mean free path of the conduction e^-/h^+; so, the mobility ↓ with ↑ T

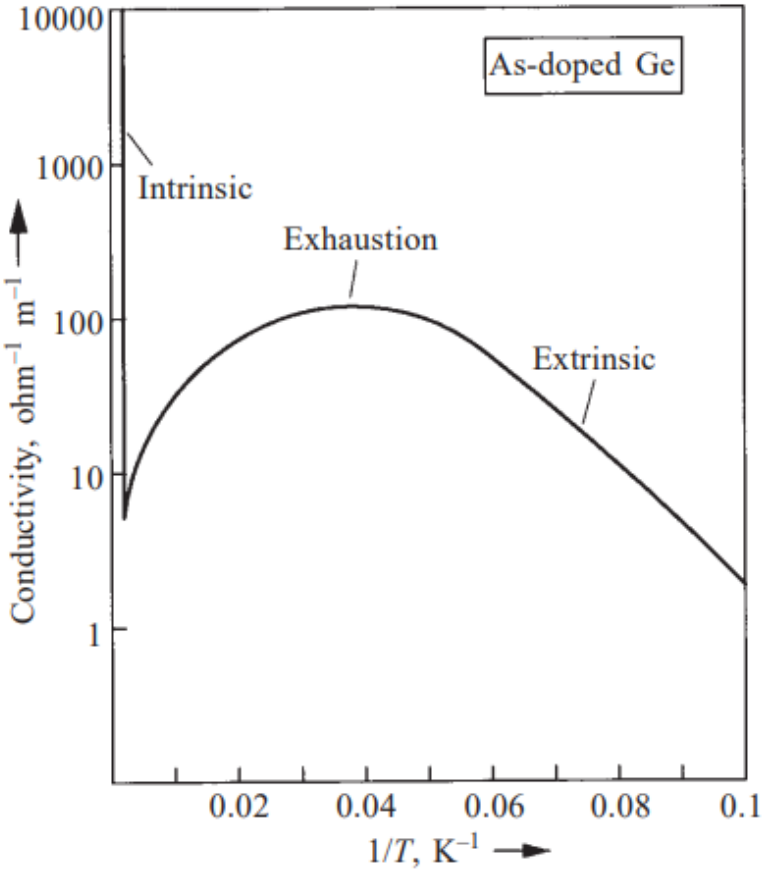


Fig. 18 The variation of log of conductivity with the reciprocal of temperature for an extrinsic semiconductor

Electrical & Electronic Properties

➢ Extrinsic Semiconductors

- Above RT, there is another linear region with a negative slope of large magnitude, which corresponds to intrinsic conduction
- The slope difference between the extrinsic and intrinsic regions reflects the difference in magnitude between the impurity ionization energy and the energy gap E_g
- For stability of operation of a semiconductor device, a relatively flat region, where the conductivity does not vary appreciably with changes in ambient temperature, is often desirable
- On the other hand, if the device is to be used as a thermistor for measuring temperature, maximum sensitivity in measurement is desirable

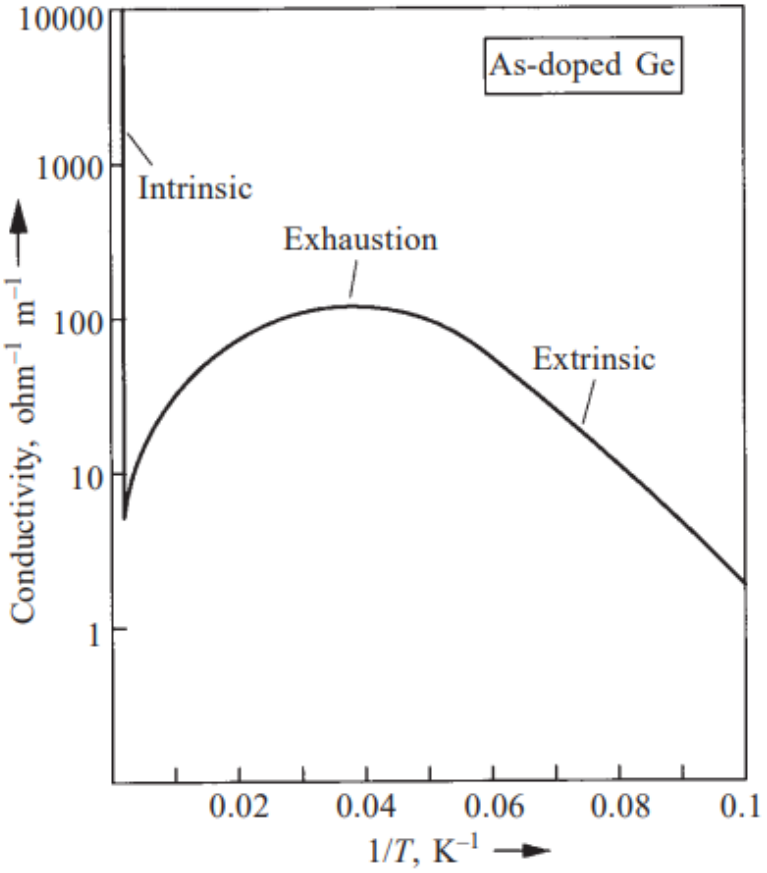


Fig. 18 The variation of log of conductivity with the reciprocal of temperature for an extrinsic semiconductor

Electrical & Electronic Properties

- **Doping**

- Impurity doping can be carried out by solid state diffusion under two conditions

- A constant surface concentration of the dopant is maintained at a high enough temperature so that the dopant diffuses in and creates a characteristic concentration profile

- A constant total quantity of the dopant is first deposited on the surface (predeposition) either by short-time annealing or by other techniques such as ion implantation at a low temperature

- The concentration profile is then altered through a high temperature anneal in a non-doping atmosphere (drive-in)

- A p-n junction is created by diffusing into the bulk semiconductor through a window in the oxide layer an impurity, say, a p-type, where the bulk crystal has already been doped to be the n-type

Electrical & Electronic Properties

➢ Junction Diodes

- When a pure semiconductor is doped to become n-type, the Fermi level shifts up from the middle of the energy gap towards the donor level
- This is so because the position corresponding to 50% probability of occupation moves up due to the relatively high concentration of donor electrons in the conduction band
- If the crystal is p-type, the Fermi level shifts down towards the acceptor level
- When the same crystal is doped to become n-type on one side and p-type on the other side, the Fermi level has to be constant throughout the crystal in thermal equilibrium
- This results in the electron energy levels at the bottom of the conduction band in the n-part to be lower than those in the p-part, by an amount equal to the contact potential eV_0

Electrical & Electronic Properties

- **Junction Diodes**

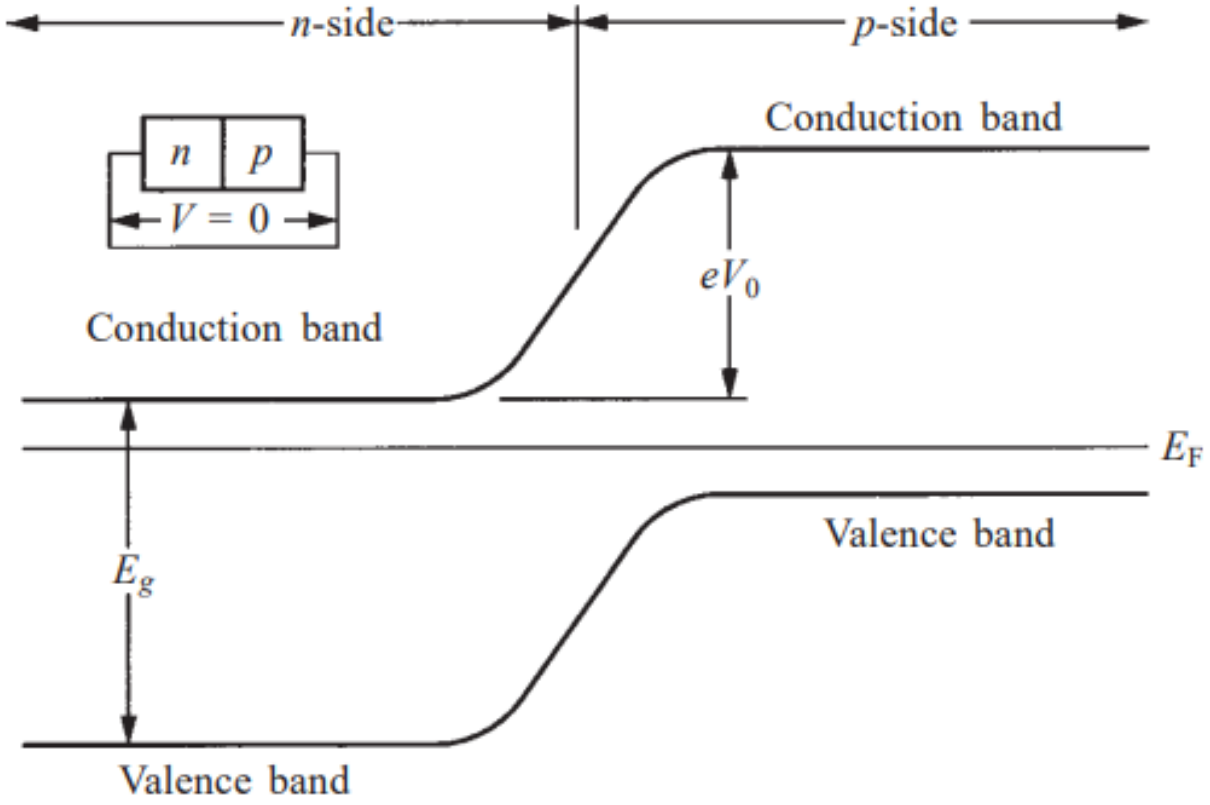


Fig. 19 The Fermi level is the same on both sides of a p-n junction. The contact potential at the junction gives rise to an energy barrier eV_0

Electrical & Electronic Properties

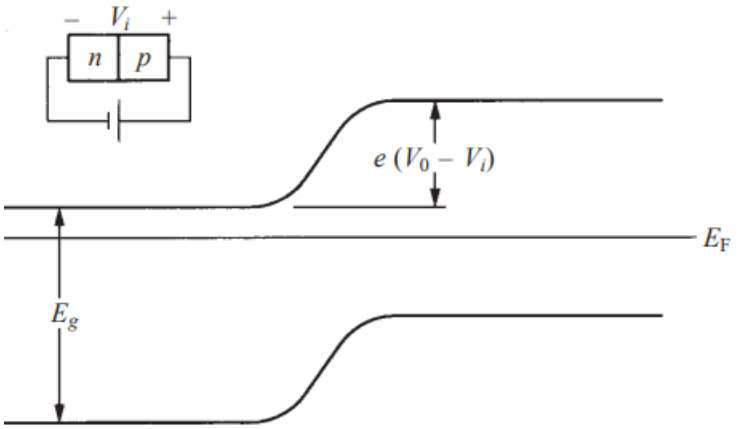
Junction Diodes

- At equilibrium, there is no net current flowing across the p-n junction
- The concentration of electrons in the conduction band on the p-side is small; these electrons can accelerate down the potential hill across the junction to the n-side, resulting in a current I_0 which is proportional to their number
- The concentration of electrons in the conduction band on the n-side is large in comparison, due to the donor contribution; however, only a small number of these electrons can flow to the p-side across the junction as they face a potential barrier
- This small number can be computed using the Boltzmann probability equation
- At equilibrium, the current from the n-side to the p-side and the current in the opposite direction, from the p-side to the n-side, are the same, equal to I_0
- There is an equal additional contribution to I_0 from the flow of holes across the junction
- The concentration of holes is large in the p-region, as compared to the n-region

Electrical & Electronic Properties

Junction Diodes

- If an external voltage V_i is now applied to the crystal such that the p-side becomes positive with respect to the n-side, the electron energy levels will change
- Note that the potential for electrons is opposite in sense to the conventional method of representing the sign of an electric potential
- The barrier at the junction is now lowered by an amount eV_i, resulting in a greatly enhanced current flow in the forward direction, i.e. from n-side to p-side
- This change in barrier does not affect the flow of electrons in the reverse direction, from the p-side to the n-side, as the flow here is still down the potential hill.
- So, the applied voltage causes a large net current flow in the forward direction

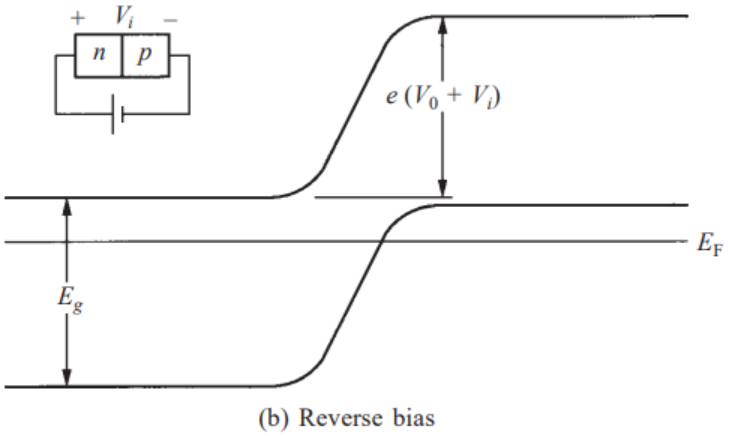


(a) Forward bias

Electrical & Electronic Properties

➢ Junction Diodes

- If an external voltage V_i is now applied in the reverse direction, the potential barrier for electrons at the junction is increased by an amount eV_i
- This would drastically reduce the current flow from the n-side to the p-side
- In the reverse direction, the flow remains unchanged at I_0



(b) Reverse bias

$$I_{\text{forward}} = I_0 \left[\exp\left(\frac{eV_i}{kT}\right) - 1 \right] \approx I_0 \exp\left(\frac{eV_i}{kT}\right)$$

$$I_{\text{reverse}} = I_0 \left[1 - \exp\left(-\frac{eV_i}{kT}\right) \right] \approx I_0$$